Activity-4 Hashing problem.

Primary hash function is h(x) = x mod 10

Input order = 4371, 1323, 6173, 4199, 4344, 9679, 1989.

initial hash values

Key	x mod 10
4371	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

a. Separate chaining

Index	Chain
0	-
1	4371
2	-
3	1323 → 6173
4	4344
5	-
6	-
7	-
8	
9	4199 → 9679 → 1989

b. Open Addressing linear probing.

Formula: $h_i(x) = (h(x) + i) \bmod 10$

4371 → index 1

1323 → index 3

6173 → 3 occupied → 4,

4199 → 9

4344 → 4 occupied → 5

9679 → 9 occupied → 0

1989 → 9 occupied → 10

Final table:

Index	Value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

C. Quadratic probing

use

$h_i(x) = (h(x) + i^2) \bmod 10$

4371 → 1

1323 → 3

6173 → 3 → occupied $3 + 1^2 = 4$

4199 → 9

4344 → 4 occ → $4 + 1^2 = 5$

9679 → 9 occu → $9 + 1^2 = 0$

1989 → 9 occ

$9 + 1 = 0$, occ

$9 + 4 = 3$

$9 + 9 = 8$, empty → 8

Final table

Index	Value
0	9679
1	4371
2	–
3	1323
4	6173
5	4344
6	–
7	–
8	1989
9	4199

D. Double hashing

Second:

$h_2(x) = 7 - (x \bmod 7)$

probe formula

$h_i(x) = (h(x) + i \cdot h_2(x)) \bmod 10$

Insertions

4371:

h = 1 → index 1

1323:

h = 1 index 3

6173 h = 3 collision

3 + 5 = 8

4199 h = 9 → index 9

4344 · h = 4 → index 4

9679: h = 9 collis.

9679 mod 7 = 5 index 5

Final double hashing table

Index	Value
0	–
1	4371
2	–
3	1323
4	4344
5	9679
6	–
7	–
8	6173
9	4199